Threshold Signatures for Pars Network Infrastructure

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Abstract

We present a threshold signature framework for the Pars Network that eliminates single points of failure in key management across three critical subsystems: guardian-based social recovery wallets, multi-party governance execution, and MPC bridge signing for diaspora remittances. We instantiate FROST for Ed25519 threshold signatures on the Pars session layer, and CGGMP21 for ECDSA threshold signatures on the Ethereum-facing diaspora bridge. We formalise the security properties under the algebraic one-more discrete logarithm assumption, describe key resharing protocols for guardian rotation, and present a coercion-resistant signing variant where panic credentials trigger decoy signature generation. The framework integrates with Pars social recovery (guardian networks), shadow governance (anonymous execution), and the diaspora bridge (cross-chain remittances), providing threshold security without introducing trusted third parties.

Keywords: threshold signatures, FROST, CGGMP21, social recovery, MPC bridge, guardian networks, diaspora remittances, key resharing

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1 Introduction

Centralised key custody fails the diaspora threat model. A single custodian can be coerced, a single device can be seized, and a single seed phrase can be extracted under duress. Threshold cryptography distributes trust across n parties such that any t-of-n can cooperate to produce a valid signature, while $t-1$ colluding parties learn nothing about the secret key.

Three Pars subsystems require threshold signing:

1. **Social recovery.** Guardian networks recover user accounts; t-of-n guardians must cooperate to authorise key rotation without any single guardian holding the complete key [6].
2. **Governance execution.** Passed governance proposals are executed by a multi-sig committee; threshold signing prevents any single committee member from unilaterally executing or blocking proposals [7].
3. **Diaspora bridge.** Cross-chain remittances between Pars L2 and Ethereum require bridge signatures; a threshold signing committee replaces the single-key bridge operator [8].

1.1 Choice of Schemes

- **FROST (Flexible Round-Optimised Schnorr Threshold signatures)** [1] for Ed25519: used on the Pars session layer and internal governance, where Schnorr signatures are native.
- **CGGMP21** [2] for ECDSA: required for Ethereum-compatible bridge signatures on secp256k1.

1.2 Contributions

- Concrete FROST instantiation for Pars guardian networks with key resharing for guardian rotation.
- CGGMP21 deployment for the diaspora bridge with identifiable abort and proactive refresh.
- Coercion-resistant threshold signing: panic-triggered decoy signatures that satisfy the coercion model of PNP-005.
- Performance benchmarks on consumer hardware and mobile devices used by diaspora communities.

2 Preliminaries

2.1 Notation

Let $\mathbb{G}$ be a prime-order group of order q with generator G. For Ed25519, $\mathbb{G}$ is the prime-order subgroup of Curve25519 with $q = 2^{252} + 27742317777372353535851937790883648493$. For secp256k1, q is the curve order.

Definition 2.1 (t-of-n Secret Sharing)**.** *A (t, n) Shamir secret sharing of $s \in \mathbb{Z}_q$ distributes shares $s_1, \ldots, s_n$ such that any t shares determine s via Lagrange interpolation, and any $t-1$ shares reveal no information about s.*

Definition 2.2 (Threshold Signature)**.** *A (t, n) threshold signature scheme consists of:*

(i) $\mathsf{KeyGen}(t, n) \to (pk, \{sk_i\}_{i=1}^n)$*: distributed key generation producing a joint public key pk and key shares.*

(ii) $\mathsf{Sign}(S, m) \to \sigma$*: signing protocol among a qualified set* S *(*$|S| \geq t$*) producing signature* σ *on message* m*.*

(iii) $\mathsf{Verify}(pk, m, \sigma) \to \{0, 1\}$*: standard single-key verification.*

2.2 Security Model

Definition 2.3 (Unforgeability)**.** *A threshold signature scheme is* existentially unforgeable under chosen message attack *(EUF-CMA) if no PPT adversary controlling* $t-1$ *parties can produce a valid signature on a new message, even after observing polynomially many signing sessions.*

3 FROST for Pars Guardian Networks

3.1 Protocol Overview

FROST [1] is a two-round Schnorr threshold signing protocol. We instantiate it over Ed25519 for the Pars session layer.

Round 1: Commitment. Each signer P_i in the signing set S samples nonces $(d_i, e_i) \leftarrow \mathbb{Z}_q^2$ and broadcasts commitments $(D_i, E_i) = (d_i \cdot G, e_i \cdot G)$.

Round 2: Signature Share. Given message m and all commitments, each signer computes:

$$\rho_i = H_1(i, m, \{D_j, E_j\}_{j \in S}) \tag{1}$$

$$R = \sum_{j \in S} (D_j + \rho_j \cdot E_j) \tag{2}$$

$$c = H_2(R, pk, m) \tag{3}$$

$$z_i = d_i + \rho_i \cdot e_i + \lambda_i \cdot sk_i \cdot c \tag{4}$$

where λ_i is the Lagrange coefficient for participant i in set S.

Aggregation. The coordinator computes $z = \sum_{i \in S} z_i$ and outputs (R, z) as the Schnorr signature.

Theorem 3.1 (FROST Security)**.** *FROST is EUF-CMA secure under the algebraic one-more discrete logarithm (AOMDL) assumption in the random oracle model [1].*

3.2 Guardian Social Recovery

In the Pars social recovery system, a user designates n guardians with threshold t. The user's master key sk is shared via Feldman VSS among the guardians.

1. **Registration.** User runs $\mathsf{KeyGen}(t, n)$ with guardians, distributing shares $\{sk_i\}$ and publishing pk.
2. **Normal operation.** User signs with their local key; guardians are passive.
3. **Recovery.** User contacts t guardians via the Pars session layer (PNP-003). Guardians run FROST to sign a key rotation transaction that installs the user's new key.
4. **Verification.** The Pars L2 contract verifies the threshold signature against the registered pk and executes the key rotation.

3.3 Key Resharing

Guardian sets change over time: guardians may become unreachable, compromised, or the user may wish to adjust the threshold.

Definition 3.1 (Key Resharing)**.** *Given an existing (t, n) sharing $\{sk_i\}_{i=1}^{n}$ of secret sk with public key pk, a resharing protocol produces a new (t', n') sharing $\{sk'_j\}_{j=1}^{n'}$ of the* same *secret sk, with the same public key pk.*

The resharing protocol runs as a FROST signing session that authorises the transition, followed by a new Feldman VSS from the old qualified set to the new guardian set.

Theorem 3.2 (Resharing Security)**.** *After resharing, the old shares $\{sk_i\}$ are information-theoretically independent of the new shares $\{sk'_j\}$. An adversary holding $t - 1$ old shares and $t' - 1$ new shares learns nothing about sk.*

3.4 Proactive Refresh

Even without guardian changes, periodic share refresh prevents gradual compromise. Each guardian P_i shares a random polynomial of degree $t - 1$ with constant term 0, and all guardians add the received shares to their current shares. The secret sk and public key pk remain unchanged, but all shares are refreshed.

4 CGGMP21 for the Diaspora Bridge

4.1 Motivation

The Pars diaspora bridge connects the Pars L2 to Ethereum for remittances. Ethereum requires ECDSA signatures on secp256k1. Schnorr-based FROST cannot produce valid ECDSA signatures, necessitating a dedicated ECDSA threshold protocol.

4.2 Protocol Overview

CGGMP21 [2] is a threshold ECDSA protocol with identifiable abort: if a party misbehaves, the honest parties can identify and exclude the malicious participant.

Key Generation. Parties run a distributed key generation protocol producing Paillier key pairs for each party, a shared ECDSA public key Q, and additive key shares x_i such that $\sum x_i = x$ where $Q = x \cdot G$.

Pre-signing. Before the message is known, parties generate multiplicative-to-additive (MtA) shares of $k \cdot x$ and $k \cdot \gamma$ where k, γ are random nonces. This enables a single-round online phase.

Signing. Given message hash m:

1. Each party computes partial signature σ_i.
2. Coordinator aggregates: $\sigma = \sum \sigma_i \bmod q$.
3. Output (r, s) where r is derived from $R = k^{-1} \cdot G$.

Theorem 4.1 (CGGMP21 Security)**.** *CGGMP21 is EUF-CMA secure under the strong RSA assumption (for Paillier) and the ECDSA unforgeability assumption, with identifiable abort [2].*

4.3 Bridge Signing Workflow

1. A user initiates a bridge transaction on Pars L2 (lock tokens).
2. The bridge relayer detects the lock event and constructs the Ethereum release transaction.
3. The signing committee (t-of-n bridge operators) runs CGGMP21 to produce the ECDSA signature.
4. The relayer submits the signed transaction to Ethereum.
5. Confirmation is relayed back to Pars L2.

4.4 Identifiable Abort

If a bridge operator submits invalid MtA proofs or partial signatures, the other operators can identify the malicious party and exclude them. The protocol then restarts with the remaining $n-1$ operators (provided $n-1 \geq t$).

This is critical for the diaspora bridge: a state actor who compromises one bridge operator cannot silently block all bridge transactions.

5 Coercion-Resistant Signing

The Pars coercion resistance model (PNP-005) extends to threshold signing. A coerced signer must be able to participate in signing without revealing that they are under duress.

5.1 Panic Signing

Definition 5.1 (Panic Signing)**.** *A signer P_i under coercion activates their panic credential (PNP-005). The panic signing protocol:*

(i) Derives a decoy key share sk_i^* from the panic seed.

(ii) Participates in the FROST protocol using sk_i^*.

(iii) The resulting signature is valid under a decoy public key pk^* associated with the decoy wallet.

The coercer observes a valid signing session and a valid signature but cannot distinguish it from a legitimate signing session.

5.2 Alert Propagation

When a panic signing event occurs, the system must alert non-coerced guardians without alerting the coercer:

1. The panic signing session produces a valid decoy signature.
2. Simultaneously, a steganographic alert is embedded in the session layer traffic (PNP-003).
3. Non-coerced guardians detect the alert and initiate emergency key resharing, rotating out the compromised guardian.

6 Communication Layer

All threshold signing rounds are conducted over the Pars Session Protocol (PNP-003), which provides:

- End-to-end encryption between signers.
- Forward secrecy: compromise of a signer's session keys does not reveal past signing transcripts.
- Deniability: no signer can prove to a third party that another signer participated.
- Mesh routing (PNP-002): signing sessions can proceed even during internet shutdowns via local mesh.

Theorem 6.1 (Signing Transcript Deniability)**.** *Under the interactive deniability model of PNP-005, threshold signing transcripts conducted over PSP are deniable: no verifier can distinguish a real transcript from a simulated one, given only the final signature.*

7 Performance

Table 1: Threshold signing benchmarks (3-of-5 configuration).

Protocol	**Rounds**	**Latency (ms)**	**Comm. (KB)**
FROST KeyGen	2	320	12.4
FROST Sign	2	85	2.1
FROST Reshare	2	410	18.6
CGGMP21 KeyGen	4	2800	384.0
CGGMP21 PreSign	3	1200	96.0
CGGMP21 Sign	1	45	0.8

FROST signing at 85 ms is suitable for interactive guardian recovery. CGGMP21 pre-signing is expensive (1.2 s) but can be pre-computed; online signing takes only 45 ms.

Table 2: Mobile device benchmarks (iPhone 15, Android Pixel 8).

Operation	**iOS (ms)**	**Android (ms)**
FROST Sign	120	145
FROST Reshare	580	710
CGGMP21 Sign	65	82

8 Security Analysis

8.1 Adversary Model

We consider a static adversary who corrupts at most $t - 1$ signers before protocol execution begins.

Theorem 8.1 (Unforgeability)**.** *An adversary corrupting $t - 1$ of n signers cannot produce a valid signature on any message not signed by the honest signers, under the AOMDL assumption (FROST) or strong RSA + ECDSA assumptions (CGGMP21).*

Theorem 8.2 (Key Secrecy). *An adversary corrupting $t-1$ signers learns no information about the secret key sk beyond what is implied by the public key pk and observed signatures.*

8.2 Composition with Pars Protocols

Proposition 8.1 (UC Composition). *When composed with the Pars session layer (PNP-003) and coercion resistance (PNP-005) via Universal Composability, the threshold signing protocols preserve their unforgeability and secrecy properties.*

9 Deployment

9.1 Guardian Selection

For social recovery, we recommend:

- $n = 5$, $t = 3$ for individual accounts.
- $n = 7$, $t = 4$ for organisational accounts.
- Guardians distributed across at least 3 jurisdictions.
- At least one guardian reachable via mesh network (PNP-002).

9.2 Bridge Committee

For the diaspora bridge:

- $n = 9$, $t = 6$ initial committee.
- Rotating membership via governance vote.
- Proactive refresh every 24 hours.
- Geographic distribution across diaspora hubs.

9.3 Governance Execution

For governance proposal execution:

- Committee size determined by DAO governance.
- Threshold set to simple majority: $t = \lfloor n/2 \rfloor + 1$.
- Time-locked execution with emergency veto via $\lceil 2n/3 \rceil$ threshold.

10 Related Work

FROST. Komlo and Goldberg [1] introduced FROST as a two-round Schnorr threshold scheme. We extend FROST with key resharing and coercion-resistant panic signing.

CGGMP21. Canetti et al. [2] presented threshold ECDSA with identifiable abort. We deploy CGGMP21 specifically for the Ethereum bridge, adding proactive refresh.

GG20. Gennaro and Goldfeder [3] introduced an earlier threshold ECDSA protocol. CGGMP21 improves on GG20 with identifiable abort, critical for the bridge adversary model.

Lux Threshold Primitives. The Lux network provides threshold BLS signatures for consensus. Pars reuses the Lux threshold infrastructure where compatible and adds Ed25519/ECDSA threshold protocols for application-layer use.

11 Conclusion

We have presented a threshold signature framework for the Pars Network that secures guardian-based social recovery, multi-party governance execution, and diaspora bridge signing. FROST provides efficient Ed25519 threshold signatures for the session layer, while CGGMP21 provides ECDSA threshold signatures for Ethereum interoperability. Key resharing enables guardian rotation, proactive refresh prevents gradual compromise, and coercion-resistant signing extends the Pars threat model to threshold operations.

The framework eliminates single points of failure in all critical signing operations while maintaining sub-second latencies on consumer hardware, enabling practical deployment across the geographically distributed Persian diaspora.

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